

Liquid TQFT Formalization Project: Progress Report and Contributor Roadmap

Status: Active | Lean 4 + Mathlib v4.28.0 | March 2026 Sessions: 6 Aristotle + Claude synthesis | Files: 6 Lean | ~2200 lines

Project Overview

This project investigates whether liquid vector spaces (Clausen-Scholze) can serve as a target category for topological quantum field theories, replacing topological vector spaces which fail to be abelian. The investigation is conducted via formal verification in Lean 4 against Mathlib, assisted by Harmonic’s Aristotle system.

The core claim: condensed abelian groups form a symmetric monoidal abelian category with exact tensor products, making them categorically suitable as TQFT targets where TopVect is not.

Completed Work

Session 1: LiquidTQFT.lean (Abstract Framework)

Result	Status	Notes
<code>AbstractTQFT</code> structure (braided monoidal functor)	Sorry-free	
<code>AbstractTQFT.transfer</code> (composition preserves TQFT)	Sorry-free	The key theorem
<code>Abelian CondensedAb</code>	Sorry-free	<code>inferInstance</code>
<code>abelian_mono_epi_is_iso</code> (key <code>TopAb</code> obstruction)	Sorry-free	
<code>HasKernels</code> , <code>HasCokernels</code> , <code>finite (co)limits</code>	Sorry-free	<code>inferInstance</code>
<code>MonoidalCategory CondensedAb</code>	Sorry-free	Patched post-Session 3

<code>BraidedCategory CondensedAb</code>	Sorry-free	Patched post-Session 3
<code>SymmetricCategory CondensedAb</code>	Sorry-free	Patched post-Session 3
Gluing structure from exactness	Structural	Takes exactness as hypothesis

Note: The monoidal instances were originally axioms (Session 1), replaced with proved instances after Session 3.

Session 2: CondensedMonoidal.lean (Manual Construction, Superseded)

Result	Status	Notes
Full <code>MonoidalCategoryStruct</code> on <code>CondensedAb</code>	Sorry-free	All data defined
<code>MonoidalCategory</code> axioms (5/10)	5 sorry	Blocked by 2 comparison isos
Key blocker analysis	Complete	Led to Session 3

Superseded by `MonoidalViaLocalization.lean`. Retained for reference and its analysis of the sheafification comparison problem.

Session 3: MonoidalViaLocalization.lean (The Breakthrough)

Result	Status	Notes
<code>condensedAb_W_isMonoidal</code>	Sorry-free	<code>GrothendieckTopology.W.monoidal</code>
<code>MonoidalCategory CondensedAb</code>	Sorry-free	6 lines of Lean
<code>BraidedCategory CondensedAb</code>	Sorry-free	
<code>SymmetricCategory CondensedAb</code>	Sorry-free	
<code>presheafToSheaf</code> is monoidal and braided	Sorry-free	<code>inferInstance</code>

Key discovery: `Mathlib.CategoryTheory.Sites.Monoidal` already had everything. The proof uses the internal hom (monoidal closed structure) rather than stalks.

Session 4: BanachEmbedding.lean (First Bridge)

--	--	--

Result	Status	Notes
Mathlib audit (C(S,V) structure, condensed infra, SemiNormedGrp)	Sorry-free	15+ inferInstance checks
Universe bridging instances	Sorry-free	hasLimitsOfSizeShrink
<code>banachPresheaf V (presheaf S -> C(S,V))</code>	Sorry-free	
<code>banachPresheafForgetIso</code> (link to yonedaPresheaf)	Sorry-free	
<code>PreservesFiniteProducts</code> (banachPresheaf V)	Sorry-free	Via NatIso transfer
<code>banachPresheaf_equalizerCondition</code>	Sorry-free	Via forget + isSheaf transfer
<code>banachCondensed V : CondensedAb</code>	Sorry-free	
<code>banachPresheafMap</code> (postcomposition)	Sorry-free	
<code>semiNormedGrpToCondensedAb : SemiNormedGrp -> CondensedAb</code>	Sorry-free	

Key discovery: Zero existing connection between `Analysis.NormedSpace` and `Condensed` in Mathlib. This file creates the first bridge.

Session 5: BanachEmbedding.lean (Properties + Mathematical Correction)

Result	Status	Notes
<code>semiNormedGrpToCondensedAb_faithful</code>	Sorry-free	Constant functions separate points
<code>semiNormedGrpToCondensedAb_full</code>	DISPROVED	Counterexample found (see below)
<code>SemiNormedGrp.hasFiniteProducts</code>	Sorry-free	New Mathlib result
<code>SemiNormedGrp.piFan</code> , <code>piFanLift</code> , <code>piFanIsLimit</code>	Sorry-free	Product construction
	2 sorry	Preserves equalizers +

<code>PreservesFiniteLimits</code> reduction	stubs	products
--	-------	----------

Critical finding: Fullness is mathematically false for `SemiNormedGrp`. Counterexample: let $V = c_0(\mathbb{N}, \mathbb{Z})$ (eventually-zero integer sequences with sup norm) and $W = \mathbb{Z}$. The summation map $f(a) = \sum a_n$ is continuous and additive but not bounded ($\|e_1 + \dots + e_n\| = 1$ while $f(e_1 + \dots + e_n) = n$). Postcomposition by f defines a valid condensed morphism with no lift to `SemiNormedGrp`. Root cause: continuous additive maps between seminormed abelian groups need not be bounded. The standard “continuous linear implies bounded” proof requires scalar multiplication by a dense subfield. Fullness holds for `NormedSpace` over $\mathbb{R}$ or $\mathbb{C}$.

Session 6: Cob2.lean (Combinatorial Cobordisms)

Result	Status	Notes
<code>FrobeniusObj</code> typeclass	Sorry-free	Extends <code>Mon_</code> and <code>Comon_</code>
<code>CommFrobeniusObj</code> typeclass	Sorry-free	+ commutativity axiom
<code>CommFrobeniusAlgebra</code> bundled structure	Sorry-free	
<code>CommFrobeniusData</code> (explicit 14-field record)	Sorry-free	
<code>CommFrobeniusData.toAlgebra</code>	Sorry-free	
<code>Cob2Mor</code> (inductive morphism type)	Sorry-free	$\mu, \eta, \delta, \epsilon$ + comp, tensor, swap
<code>Cob2Rel</code> (Frobenius relations)	Sorry-free	
<code>Cob2Hom</code> (quotient morphism type)	Sorry-free	
<code>Cob2Category</code> (<code>Category N</code> instance)	Sorry-free	2d cobordism category
<code>trivialFrobenius</code> (unit object is Frobenius)	Sorry-free	All 9 axioms via coherence
<code>TQFT2d</code> + <code>TQFT2d.transfer</code>	Sorry-free	Mirrors <code>AbstractTQFT</code>

<code>CommFrobeniusData.toCob2Functor</code>	1 sorry	Case analysis on quotient
--	---------	---------------------------

Non-Lean Work

Deliverable	Status
Research proposal (docx + md)	Complete, reviewed twice
MathFlowChart (221 nodes, 491 edges)	Updated
Comprehensive gap analysis and proof strategies	Complete
6 Aristotle prompts (iteratively refined)	Complete

Current Sorry Inventory

3 sorries remain across the active codebase:

Sorry	File	Difficulty	Status
<code>preservesEqualizers</code> (embedding)	BanachEmbedding.lean	Hard	4-5 layers of limit transfer
<code>preservesFiniteProducts</code> (embedding)	BanachEmbedding.lean	Hard	ContinuousMap.piEquiv path
<code>CommFrobeniusData.toCob2Functor</code>	Cob2.lean	Tedious	Case analysis on quotient

1 disproof:

Statement	File	Significance
<code>semiNormedGrpToCondensedAb_full</code>	BanachEmbedding.lean	False for SemiNormedGrp; true for NormedSpace over valued fields

Key Mathematical Findings

1. CondensedAb monoidal structure already in Mathlib (Session 3)

The entire monoidal category structure assembles from existing components via

`Sheaf.monoidalCategory`. The proof that $J.W$ is monoidal uses the internal hom, not stalks. This is more general (works for any closed braided target) and avoids the fact that `CompHaus` with the coherent topology does not have enough points in the classical sense.

2. No bridge between normed spaces and condensed math (Session 4)

Prior to this project, Mathlib had zero connection between `Analysis.NormedSpace` and `Condensed`. The embedding functor and its sheaf condition proof are the first such bridge. The main technical obstacle was universe arithmetic.

3. Fullness fails for seminormed groups (Session 5)

The embedding is faithful but NOT full. This is a genuine mathematical finding: the condensed category sees more morphisms than the normed category because continuous additive maps need not be bounded without scalar structure. The correct source for a fully faithful embedding is `NormedSpace` over a valued field, not bare seminormed groups.

4. Finite products for `SemiNormedGrp` not previously in Mathlib (Session 5)

`SemiNormedGrp.hasFiniteProducts` was constructed from scratch. Small but useful independent Mathlib contribution.

5. Commutative Frobenius algebras formalize cleanly (Session 6)

The classification “2d TQFT = commutative Frobenius algebra” is expressed using Mathlib’s existing `Mon_` and `Comon_` infrastructure. The trivial Frobenius algebra on the monoidal unit has all 9 axioms proved by the `coherence` tactic. The 2d cobordism category exists as a quotient of free morphisms by Frobenius relations.

Roadmap for Contributors

Tier 1: Approachable (days to weeks)

1.1 Preserve finite products for the embedding. Factor through `sheafToPresheaf`, use `preservesLimit_of_evaluation`, verify pointwise via `ContinuousMap.piEquiv`, lift through `forget`. 1-2 weeks.

1.2 Preserve equalizers for the embedding. Same layer structure as 1.1; continuous maps into $\ker(f-g)$ biject with equalizing maps. 1-2 weeks.

1.3 Complete the `Cob2` functor. Case analysis showing Frobenius relations are respected by the functor action. 1 week.

1.4 Project integration. Proper module structure, imports, deduplication. A few hours.

Tier 2: Focused Projects (weeks to months)

2.1 NormedSpaceCat and fully faithful embedding. Define NormedSpaceCat R , restrict embedding, prove fullness (which holds here). 2-4 weeks.

2.2 Concrete Frobenius object in CondensedAb. Condensed integers with multiplication and diagonal. First fully instantiated 2d TQFT. 2-4 weeks.

2.3 Right exactness (cokernel preservation). Open mapping theorem enters. Specifically a Banach result. 1-2 months.

Tier 3: Infrastructure Projects (months)

3.1 Projective tensor product of normed spaces. No existing Mathlib infrastructure. 2-3 months.

3.2 The embedding is monoidal. Requires 3.1. Related to nuclearity. 1-2 months after 3.1.

3.3 Port LTE exactness to current Mathlib. 3-6 months.

Tier 4: Research Frontier (year+)

4.1 Smooth cobordism categories in Lean **4.2** Concrete Chern-Simons construction valued in CondensedAb **4.3** Factorization algebras reformulated with liquid vector spaces

File Inventory

File	Lines	Sorries	Key Content
LiquidTQFT.lean	~290	0	Abstract TQFT, transfer, monoidal CondensedAb
MonoidalViaLocalization.lean	~228	0	Symmetric monoidal via localization
CondensedMonoidal.lean	~570	5	Manual construction (superseded)
BanachEmbedding.lean	~390	2	Embedding, faithfulness, fullness disproof
Cob2.lean	~375	1	Frobenius algebras, 2d cobordisms
Active total	~1280	3	Excluding superseded CondensedMonoidal

Tooling

- **Lean 4** with **Mathlib v4.28.0**
- **Harmonic Aristotle** for proof search and Mathlib API discovery (6 sessions)

- **Claude** for synthesis, prompt engineering, mathematical analysis, documents
 - All results independently verifiable via `lake build`
-

Citation

Co-authored-by: Aristotle (Harmonic) <aristotle-harmonic@harmonic.fun>

Synthesis, prompt engineering, and mathematical analysis by Claude (Anthropic).